



Operations Research

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Exercise Sheet 1

Training Exercises

Exercise T1.1. *Mining company*

A mining company owns two different pits (or mines) where certain types of ore are mined. The mines are located in different parts of the country and have different capacities. "Breaking" results in the three types of coarse-, medium- and fine-grained ore. There is a certain demand for each type of ore.

The mining company focuses on supplying at least 12 t (tons) of coarse-grained ore, 8 t of medium-grained ore, and 24 t of fine-grained ore to a metallurgical plant each week. The operating cost for the pits is 200 GE per day for pit 1 and 160 GE for pit 2. Thereby, pit 1 produces 6 t of coarse-, 2 t of medium-, and 4 t of fine-grained ore per day, while the second pit has a daily output of 2 t of coarse-, 2 t of medium-, and 12 t of fine-grained ore.

How many days should each of the pits be worked per week to meet the company's orders in an economical manner? Formulate an LP model for the solution and graphically determine an optimal solution.

Exercise T1.2. *Study hours*

Georg studies business informatics and has to write three exams in the subjects $i = 1, \dots, 3$ at the end of the semester. There are 10 weeks left until the start of the exam period and Georg has up to 15 hours to study each week. Unfortunately, Georg realizes that his knowledge in the subjects is still limited. Without further studying he would achieve the following fictitious grades:

$\bar{F}_1$	$\bar{F}_2$	$\bar{F}_3$
8.0	3.4	6.0

Through each study session, Georg can improve his fictitious grades. For each subject i he can thereby achieve the following improvement effects v_i :

$$v_1 = \begin{cases} 0.1 & \text{for the first 30 hours} \\ 0.05 & \text{for each additional hour} \end{cases}$$

$$v_2 = \begin{cases} 0.1 & \text{for the first 6 hours} \\ 0.06 & \text{for each additional hour} \end{cases}$$

$$v_3 = \begin{cases} 0.2 & \text{for the first 5 hours} \\ 0.1 & \text{for the next 10 hours} \\ 0.05 & \text{for each additional hour} \end{cases}$$

The real final grades E_i are calculated from the fictitious grades F_i as follows:

$$E_i = \begin{cases} 5.0 & , \text{ if } F_i \geq 5.0 \\ F_i & , \text{ if } 1.0 \leq F_i \leq 5.0 \\ 1.0 & , \text{ if } F_i \leq 1.0 \end{cases}$$

George's goal is to minimize the sum of the final grades. However, it must be ensured that he passes all exams, i.e., he needs a final grade of at least 4.0 in all exams.

- a) Model this problem as a linear optimization problem.
- b) Determine an optimal solution of your LP using Gurobi. According to this, how many hours should Georg study for each subject? What would be his final grade in each subject? What is the minimum sum of his final grades?

Exercise T1.3. Vitamin tablets

A manufacturer of vitamin preparations needs raw materials $R_1, \dots, R_m$ for the production of a vitamin tablet V . The cost of the raw materials is different, that is, one unit of R_k costs c_k monetary units.

The manufacturer guarantees a minimum content g_i of vitamins V_i ($i = 1, \dots, n$) per vitamin tablet. Each raw material R_k contains a proportion a_{ik} of vitamins V_i . In addition to vitamins, each raw material contains u_k units of an undesirable substance U , which is legally limited to an amount of l .

Since the manufacturer produces many different vitamin tablets, he is interested in a generic model for calculating the minimum cost of the raw materials that takes into account the legal requirements. Design a linear program that will help you decide how many units x_k of raw materials R_k should be used to produce one vitamin tablet V . The derivation of the model must be documented in comprehensible steps.

Self-study Exercises

Exercise S1.1. LP mapping

Let there be four linear programs in algebraic representation (P_1, P_2, P_3, P_4) and another four linear programs in graphical representation (G_1, G_2, G_3, G_4) . In each graphical representation, the objective function is colored blue and passes through the optimal point x^* of the linear program, which is colored red. The admissible region is colored gray.

Assign to each linear program in algebraic representation exactly one matching linear program in graphical representation, so that both describe the same underlying linear program. You may assume that for each given linear program in algebraic representation exactly one matching linear program in graphical representation is given in the problem.

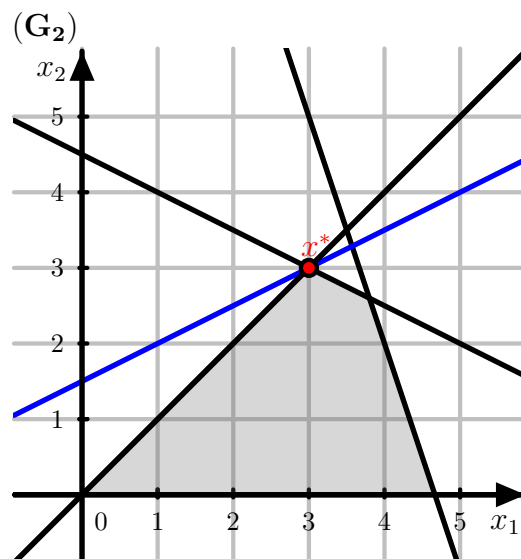
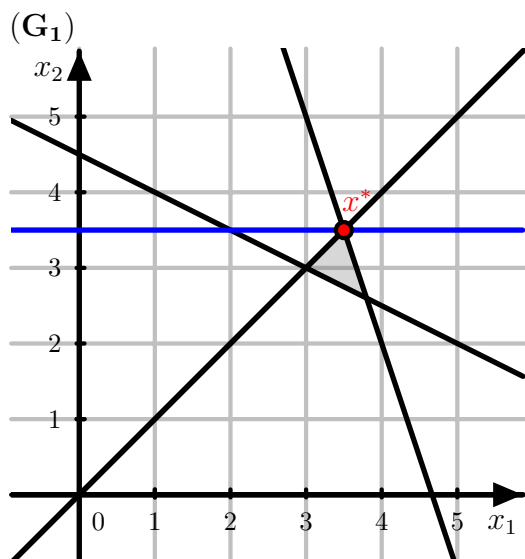
A justification for your assignment is not required.

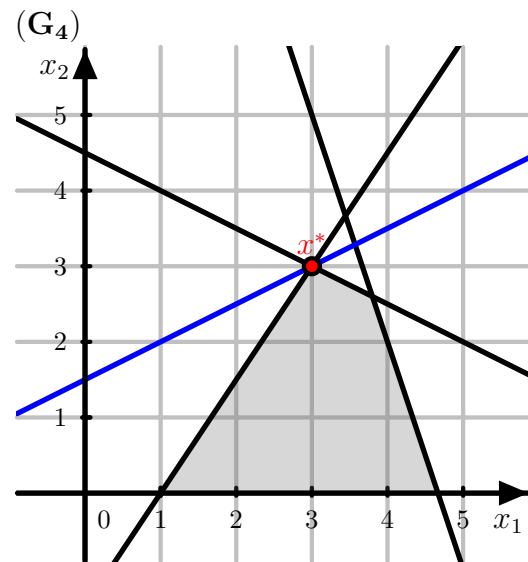
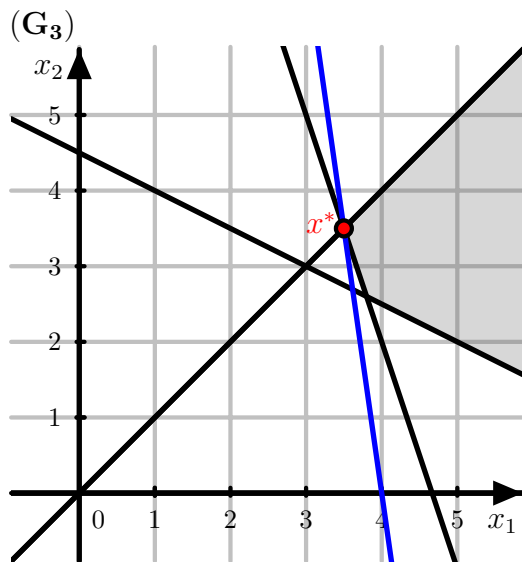
$$\begin{aligned}
 (\mathbf{P}_1) \quad & \underset{\text{s.t.}}{\text{Max}} && -x_1 + 2x_2 \\
 & x_1 + 2x_2 &\leq & 9 \\
 & 3x_1 + x_2 &\leq & 14 \\
 & -x_1 + x_2 &\leq & 0 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

$$\begin{aligned}
 (\mathbf{P}_3) \quad & \underset{\text{s.t.}}{\text{Max}} && x_2 \\
 & x_1 + 2x_2 &\geq & 9 \\
 & 3x_1 + x_2 &\leq & 14 \\
 & -x_1 + x_2 &\leq & 0 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

$$\begin{aligned}
 (\mathbf{P}_2) \quad & \underset{\text{s.t.}}{\text{Max}} && -x_1 + 2x_2 \\
 & x_1 + 2x_2 &\leq & 9 \\
 & 3x_1 + x_2 &\leq & 14 \\
 & -3x_1 + 2x_2 &\leq & -3 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

$$\begin{aligned}
 (\mathbf{P}_4) \quad & \underset{\text{s.t.}}{\text{Min}} && 7x_1 + x_2 \\
 & x_1 + 2x_2 &\geq & 9 \\
 & 3x_1 + x_2 &\geq & 14 \\
 & -x_1 + x_2 &\leq & 0 \\
 & x &\in & \mathbb{R}_{\geq 0}^2
 \end{aligned}$$





Exercise S1.2. Forecast planning

Your forecasting department provides you with the following sales forecasts, which describe how many units of your product can be sold in the first 6 months of the coming year:

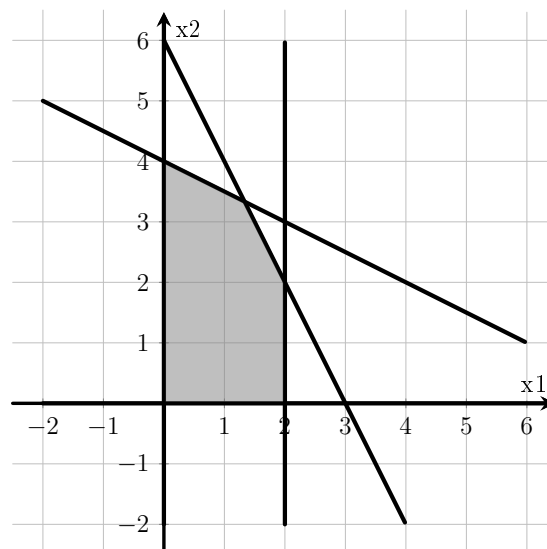
m	January	February	March	April	May	June
F_m	14,000	12,000	16,000	14,000	15,000	16,000

In December, 12,000 units were produced. If the production is increased from one month to the next, there is a cost of €2 per additional unit produced. If the production is reduced from one month to the next, there is a cost of €0.50 per fewer unit produced. Overproduced units can be stored at a cost of €1 per unit in order to be sold later. At the end of December, no units are stored and at the end of June, the warehouse should also be empty.

Using a linear program, create a production plan that is cost-minimal (in terms of production and storage costs) and that covers the forecast demand.

Exercise S1.3. Graphical solution of an LP

Given is the following feasible region (shaded in gray) of a linear problem.



- a) Specify linear constraints that define this feasible region.
- b) Specify a vector c such that $(0, 4)$ is the *only* optimal solution of the problem $\max c^T x$ subject to the constraints from part a). Draw your objective function.
- c) Consider the objective vector $c' = (-1, \beta)^T$. For which choices of $\beta \in \mathbb{R}$ is the set of optimal solutions more than a single point?